

ME 165

Basic Mechanical Engineering

Lecture 10

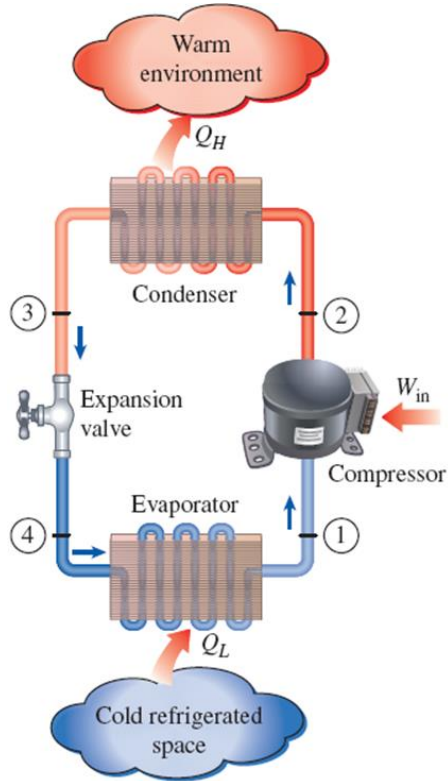
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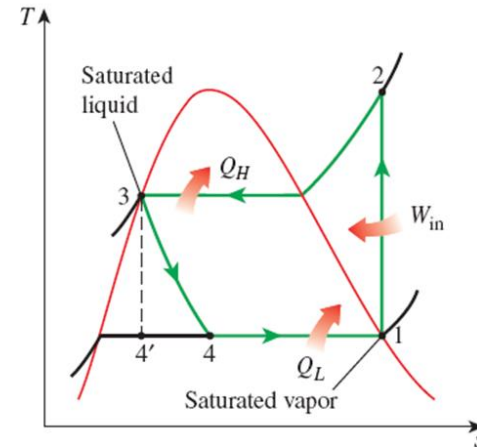


Vapor Compression Refrigeration System

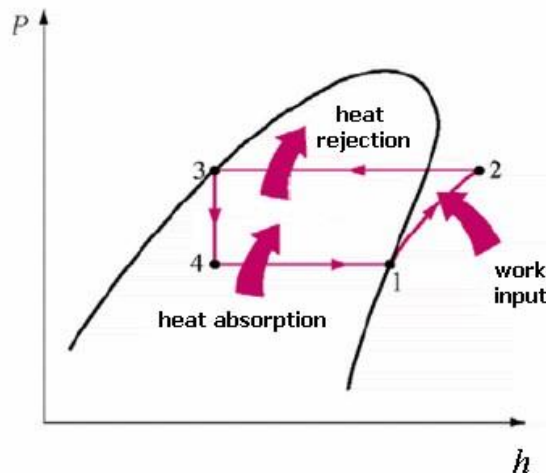
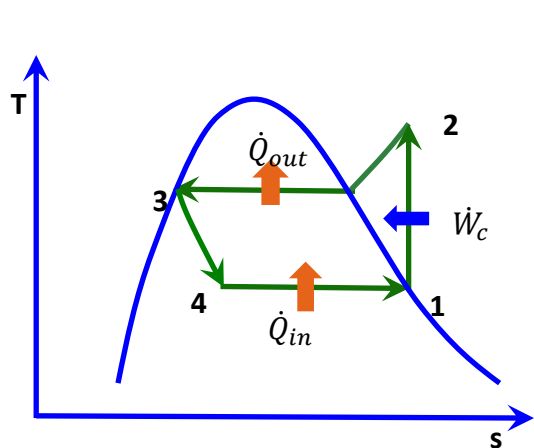


Ideal Vapor Compression Refrigeration Cycle

- 1—2 : Isentropic compression in a compressor
- 2—3 : Constant Pressure heat rejection in a condenser
- 3—4 : Throttling in an expansion device (Enthalpy constant)
- 4—1 : Constant pressure heat absorption in an evaporator



Vapor Compression Refrigeration System



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(Enthalpy constant)

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Engineering model (Assumptions):

- Each component is analyzed as a control volume at **steady state**.
- Dry compression** is presumed.
- Kinetic and potential energy changes** are ignored.

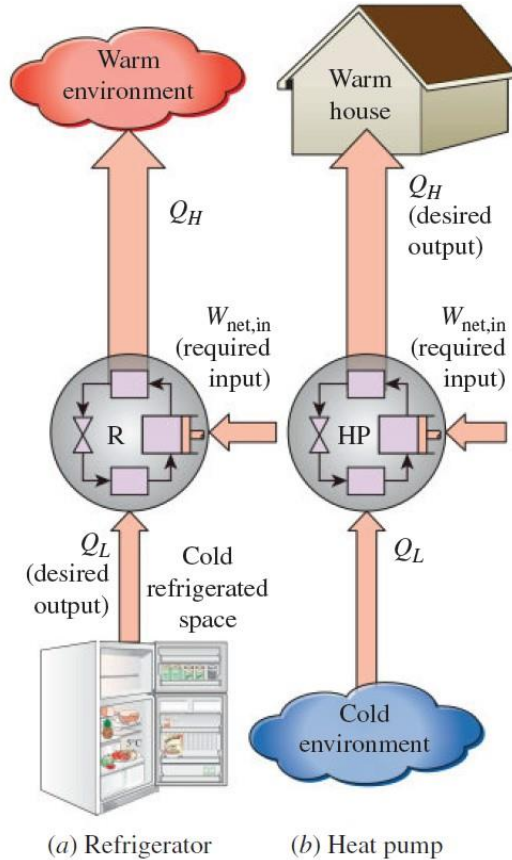
Cooling effect (Desired output), $\dot{Q}_{in} = \dot{m} (h_1 - h_4)$

Work input (Required input), $\dot{W}_c = \dot{m} (h_2 - h_1)$

$$\text{COP} = \frac{\text{Desired output}}{\text{Required input}} = \frac{\dot{Q}_{in}}{\dot{W}_c} = \frac{h_1 - h_4}{h_2 - h_1}$$



Refrigerator Vs Heat Pump



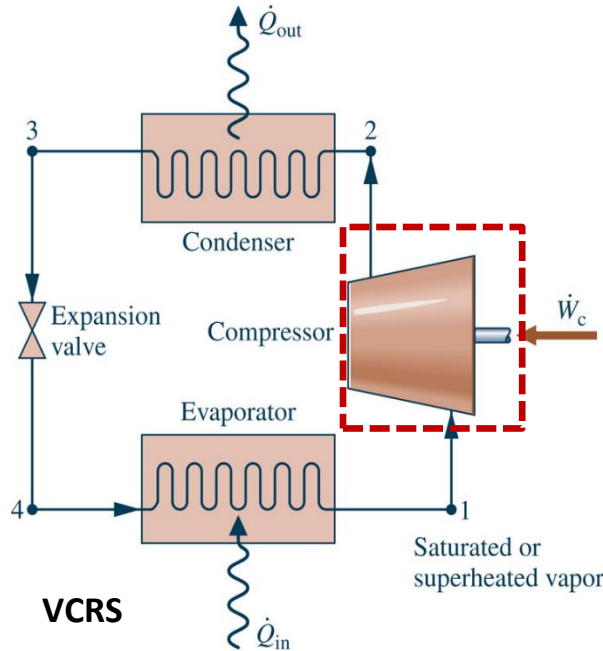
$$\begin{aligned}
 COP_R &= \frac{\text{Desired Output}}{\text{Required Input}} \\
 &= \frac{\text{Cooling effect}}{\text{Work input}} \\
 &= \frac{Q_L}{W_{net,in}}
 \end{aligned}$$

$$\begin{aligned}
 COP_{HP} &= \frac{\text{Desired Output}}{\text{Required Input}} \\
 &= \frac{\text{Heating effect}}{\text{Work input}} \\
 &= \frac{Q_H}{W_{net,in}}
 \end{aligned}$$

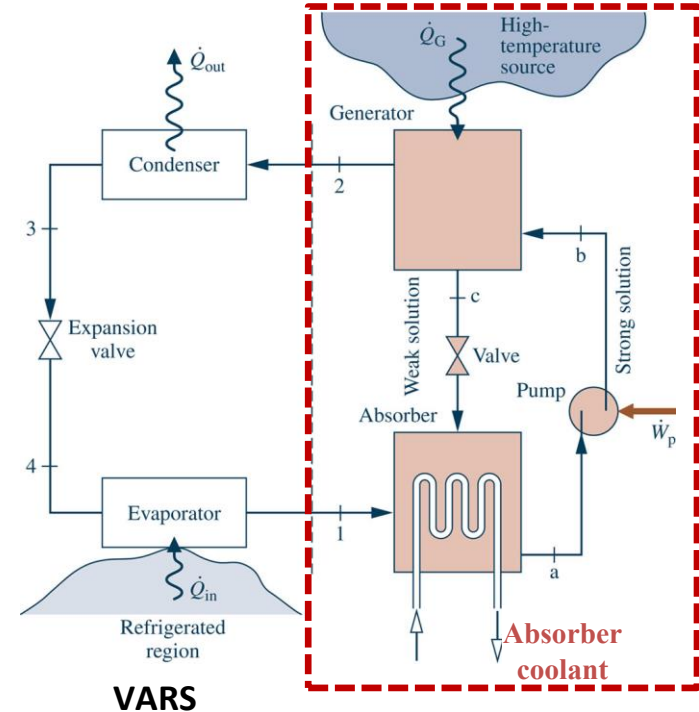
For same values of Q_H , Q_L and $W_{net,in}$

$$COP_{HP} = COP_R + 1$$

Vapor Absorption Refrigeration System



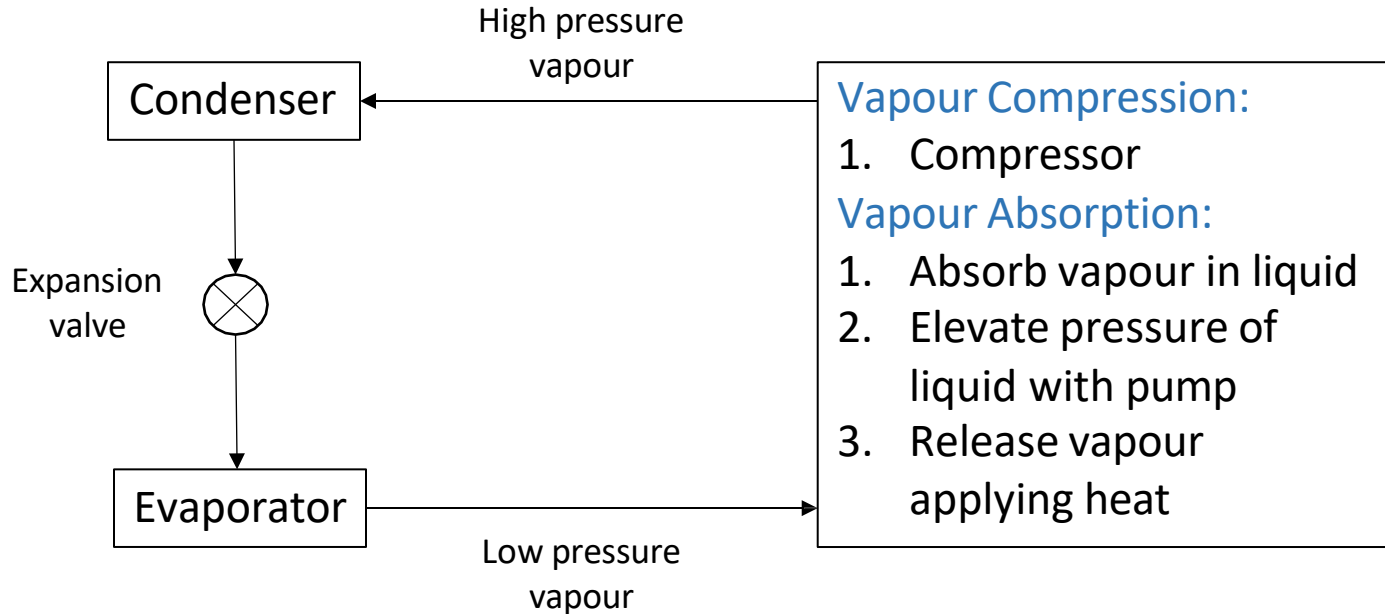
VCRS



VARS

- Absorption refrigeration systems have important commercial and industrial applications.
- Compressor is replaced by Absorber, pump and generator

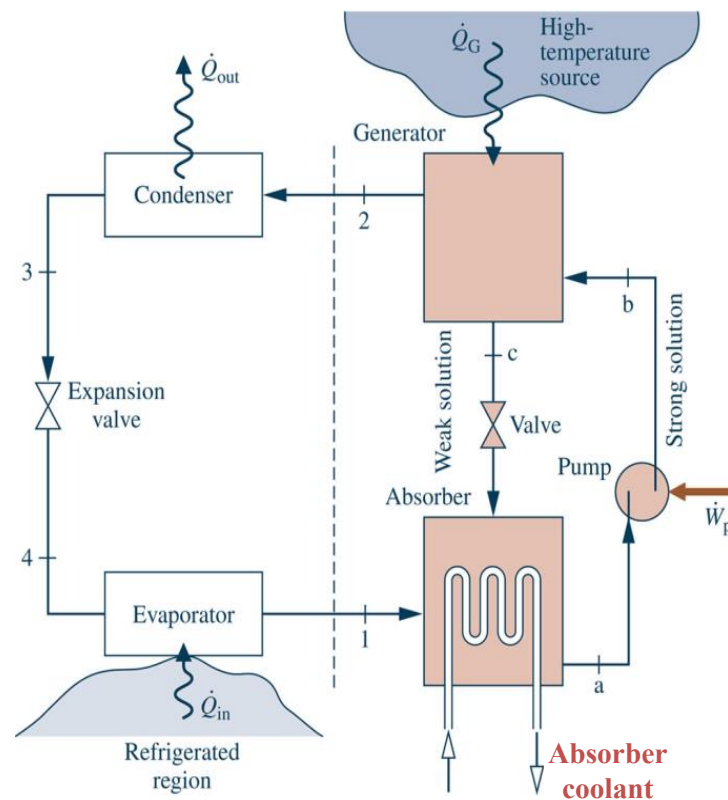
Vapor Absorption Refrigeration System



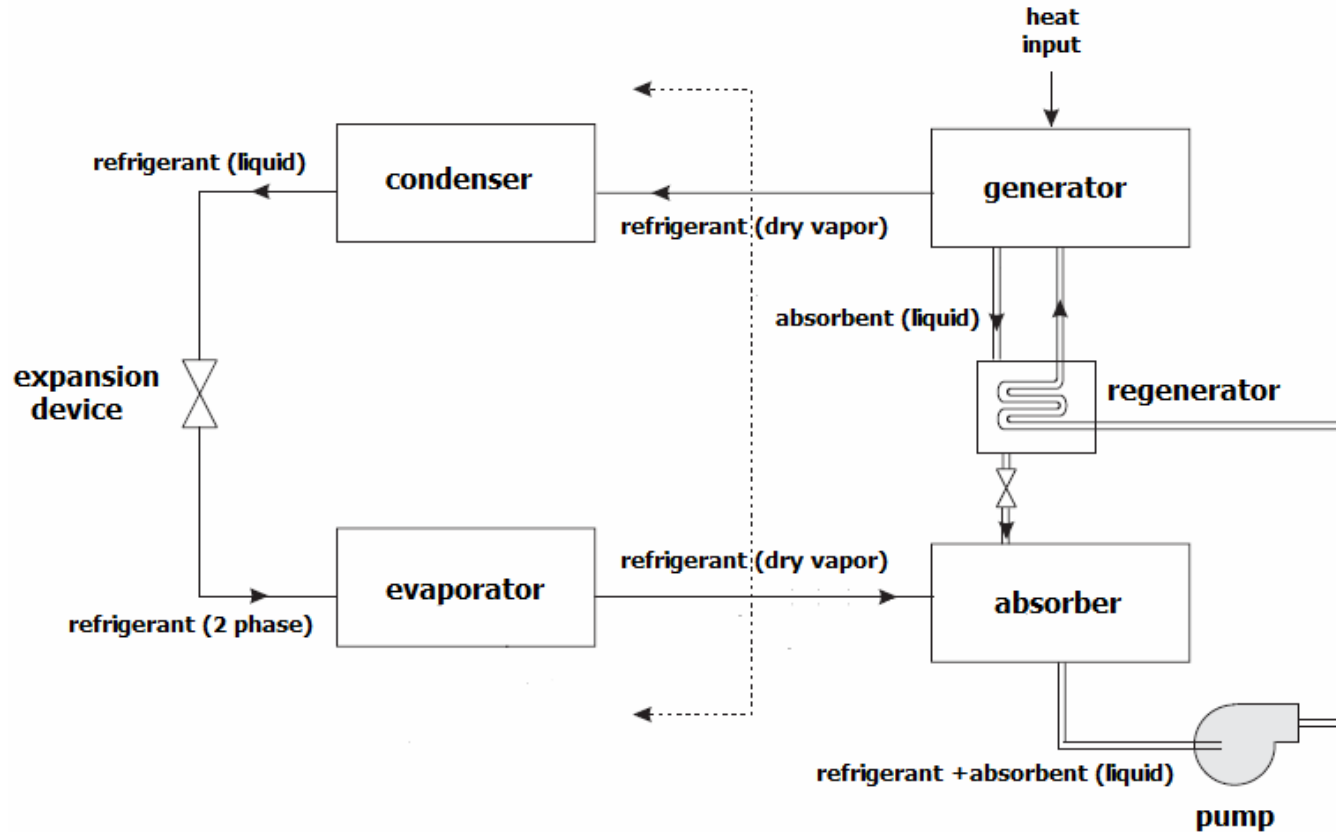
Vapor Absorption Refrigeration System

Working Principles

- ▶ Refrigerant vapor coming from the evaporator is absorbed in the absorbent to form a solution.
- ▶ The liquid solution is then **pumped** to the higher operating pressure.
- ▶ However, since only refrigerant vapor is allowed to enter the condenser, a means must be provided to retrieve refrigerant vapor from the pumped liquid solution.
- ▶ This is accomplished by the **generator** using heat transfer from a relatively high-temperature source.
- ▶ Steam or waste heat that otherwise might go unused can be a cost-effective choice for the heat transfer to the generator.
- ▶ Alternatively, the heat transfer can be provided by solar thermal energy, burning natural gas or other combustibles, and in other ways.



Vapor Absorption Refrigeration System



Vapor Absorption Refrigeration System

Process:

In a vapor absorption refrigeration system, the refrigerant circulates through the condenser, expansion valve and evaporator same as in the vapor compression refrigeration system. The compressor is replaced by an absorber, pump, generator, regenerator and a valve.

- Two working fluids are used in an absorption system: one acts as a refrigerant and the other acts as an absorbent.
- Most common pairs of working fluids are: Ammonia (R)- Water (A) and Water (R)- LiBr (A)
- In the absorber, dry refrigerant vapor is absorbed by a liquid absorbent. The process is exothermic, and the absorption is inversely proportional to the temperature i.e. the cooler the better.
- A pump raises the solution to the pressure of the generator.
- In the generator, the refrigerant is driven out of the solution by the addition of heat (endothermic reaction) and it is passed back to the condenser.
- Sometimes a regenerator is used to recover some of the thermal energy from the absorbent solution passing back to the absorber. This energy is transferred to the solution pumped to the generator. Thus, reducing the heat input required to vaporize the solution in the generator.



Vapor Absorption Refrigeration System

Advantages of Vapor Absorption Refrigeration System:

- In a vapor absorption refrigeration system, the working fluid is pumped to higher pressure as a liquid whose specific volume is less than that of a gas as in the vapor compression refrigeration system compressor. Hence the work input is much less.
- There are considerable savings in power input because a pump is used instead of a compressor. Although sometimes this is weighed off against the cost of extra hardware in a vapor absorption refrigeration system.
- In this system, heat is added to the process to retrieve the refrigerant vapor from the liquid solution. The refrigeration process is driven by heat and this thermal energy often comes from the exhaust gasses of several other systems (boiler flue gas, gas turbine exhaust or engine exhaust). Thus the system can utilize waste heat.



Unit Of Refrigeration

Ton of Refrigeration (TR):

- It's a unit of measurement for the cooling capacity of a refrigeration machine.
- The unit of refrigeration is expressed in terms of '**Ton of refrigeration**' (TR). A ton of refrigeration is defined as the 'amount of refrigeration effect produced by the uniform freezing (or melting) of 1 ton of pure ice at 0°C in 24 hours.'
- This unit was developed during the 1880s.

$$\begin{aligned} 1\text{TR} &= 12,000 \text{ BTU/h} \\ &= 200 \text{ BTU/min} \\ &= 3.516 \text{ kW} \end{aligned}$$



Refrigerants

- A refrigerant is a compound used in a refrigeration cycle that reversibly undergoes a phase change from a gas to a liquid.
- Refrigerants can be classified as below:

Type		Industrial Name	Chemical Formula
Inorganic Refrigerants		R717	NH ₃
		R744	CO ₂
Halocarbon Refrigerants	CFC	R11	CCl ₃ F
		R12	CCl ₂ F ₂
	HCFC	R22	CHClF ₂
		R134a	CF ₃ CH ₂ F
Hydrocarbon Refrigerants		R50	CH ₄
		R290	CH ₃ CH ₂ CH ₃
Azeotropes		R500	R12 – 73.8% R115 – 26.2%
		R501	R22 – 75% R12 – 25%



Refrigerants: Nomenclature

The refrigerant nomenclature system was developed by DuPont and systematically identifies the molecular structure of refrigerants made with a single halogenated hydrocarbon. The meaning of the codes is as follows:

Inorganic Refrigerants –

The digit 7 is followed by the **molecular mass** of the refrigerant.

Hydrocarbon and Halocarbon Refrigerants –

- Rightmost digit: Number of **fluorine** atoms per molecule.
- Tens digit: **One plus the number of hydrogen** atoms per molecule.
- Hundreds digit: The number of **carbon atoms minus one**. Omitted for methyl halides, which have only one carbon atom.
- Thousands digit: **Number of double bonds** in the molecule. This is omitted when zero, and in practice is **rarely used**, since most candidate compounds are **unstable**.
- Remaining bonds not accounted for are occupied by chlorine atoms.
- A suffix of a lower-case letters a, b, or c indicates increasingly unbalanced isomers (isomer - a molecule has the same number of atoms of each element, but has a different arrangement of the atoms, It has the same molecular formula as the other molecule, but with a different chemical structure)



Refrigerants



TheEngineeringMindset.com

Boiling Point:

-40.8 °C
-41.40 °F

-26.3 °C
-15.34 °F

-48.5 °C
-55.4 °F



Boiling point of Water: 100 °C
212 °F

Activate Windows
Go to Settings to activate Windows.



Different Generations of Refrigerants

1st Gen	2 nd Gen	3 rd Gen	4 th Gen	5 th Gen
1830 -1930s	1931 -1990s	1990 -2010s	2012 onwards	2020s ?
Whatever worked	Safety and Durability	Ozone Layer Protection	Global Warming	Efficiency & Trade-offs
CO ₂ NH ₃ SO ₂ HCs H ₂ O	CFCs HCFCs HFCs NH ₃ H ₂ O	HFCs HCs CO ₂ NH ₃ H ₂ O	HFOs HCFOs HCs CO ₂ NH ₃ H ₂ O ...	CO ₂ NH ₃ H ₂ O ...



Refrigerants

Desirable Properties of Refrigerants:

The important properties of refrigerants that relate to the overall performance of a refrigeration system are given below–

- A **high latent heat of vaporization** is desirable because it is usually associated with a high refrigerating effect per unit mass of refrigerant circulated.
- A **high critical temperature** is desirable as it is impossible to condense the refrigerant at a temperature above the critical, no matter how much the pressure is increased.
- A refrigerant should have a **low boiling temperature**; Otherwise it would become necessary to operate the compressor at high vacuums with the resulting lowered efficiency and capacity.
- The **freezing point** of the refrigerant should be lower than the lowest operating temperature of the cycle to prevent blockage of refrigerant pipelines.
- **Low refrigerant densities** are usually preferable, since they permit the use of small suction and discharge lines without excessive pressure drops.



Refrigerants

Desirable Properties of Refrigerants:

- A refrigerant should have **low volume per unit mass** when in gaseous state. This not only reduces the size of the equipment but also means higher compressor efficiency
- The refrigerant should have a **lower viscosity** and **high values of thermal conductivity**.
- The refrigerant should be **chemically stable** in operating conditions
- The refrigerant should have **no chemical reaction** with the lubricating oil.
- The refrigerant should be **non toxic, non corrosive and inert**.

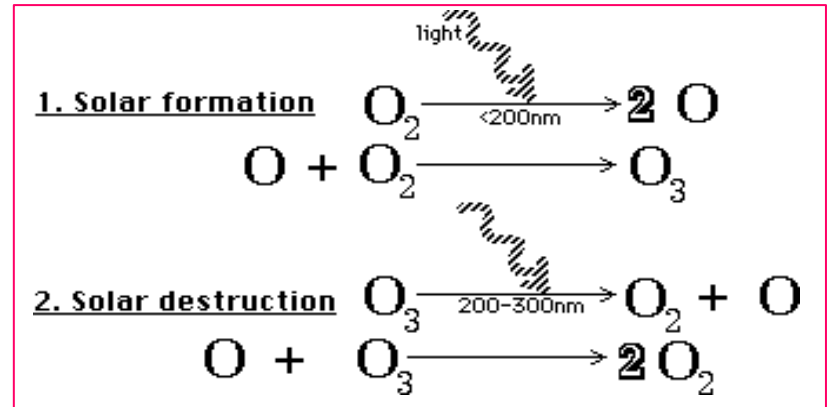


Ozone Layer Depletion

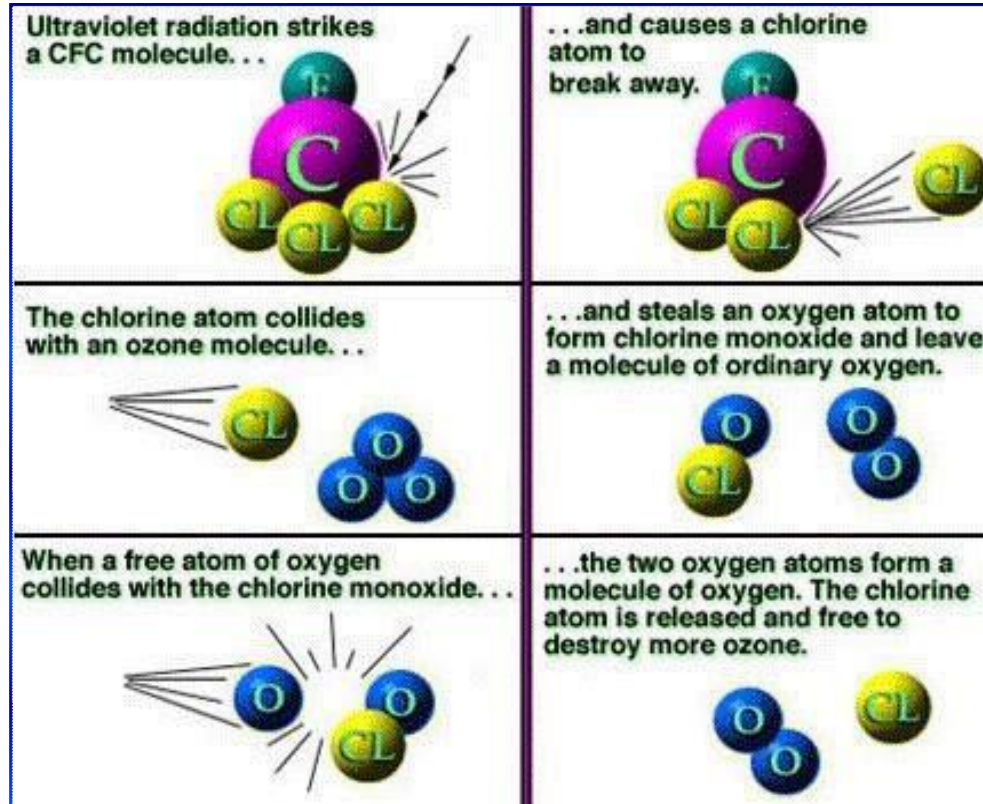
- The ozone layer is found in the lower portion of the stratosphere, approximately **15-35 km** above earth surface.
- The ozone layer contains about **10 ppm of ozone**, while the average ozone concentration in atmosphere as a whole is about 0.3 ppm of air.
- The ozone layer in the stratosphere block many of the sun's harmful rays (**UV-A and UV-B**).
- **The thicker the layer of stratospheric O₃, the greater the protection.**

Formation and Destruction:

Being a natural constituent of the stratosphere, O₃ is regularly formed and destroyed in a cyclic manner having a **dynamic steady state**.



Ozone Layer Depletion



Ozone Layer Depletion

CFC's and HCFC's deplete the O₃ layer.

- **They do not occur naturally and are totally man-made,** and are found in air conditioning and refrigeration systems, in industrial manufacturing processes, and in some cleaning solvents.
- The depleted O₃ layer allows too much ultraviolet rays down to the earth's surface, **increasing the risk of skin cancer in humans and damaging the environment,** especially harming sea life.
- O₃ depletion also leads to a **reduction in crop yield.**
- CFCs are no longer manufactured in industrialized countries, but remain in **older refrigeration equipment** including *automobile air conditioners*.
- Available illegally or in the black-markets.



Acknowledgement

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